CSES — Additional Problems I Reference

All tasks: 1 s, 512 MiB. Modulo $10^9 + 7$ only where explicitly stated.

1086 — Writing Numbers

Write $1, 2, 3, \ldots$ in decimal order. Across the whole process, each digit key $0, \ldots, 9$ may be pressed at most n times; find the largest x for which every number $1, \ldots, x$ can be written completely. **In:** `n`

Out: last writable number x

Constraints: $1 \le n \le 10^{18}$. Use 64-bit.

1087 — Shortest Subsequence

Given a DNA string over `A,C,G,T`, find a shortest DNA string that is *not* a subsequence of it. Any shortest answer is accepted. **In:** DNA string s

Out: any shortest non-subsequence

Constraints: $1 \le |s| \le 10^6$.

1142 — Advertisement

A histogram has n unit-width bars of heights k_i. Find the maximum area of an axis-aligned rectangle contained under consecutive bars. **In:** `n`; $k_1 \ldots k_n$

Out: maximum rectangle area

Constraints: $n \le 2 \cdot 10^5$, $k_i \le 10^9$. Use 64-bit.

1147 — Maximum Building I

Given an $n \times m$ grid of empty cells `.` and trees `*`, find the largest axis-aligned rectangle consisting only of empty cells. **In:** `n m`; then n grid rows

Out: maximum empty-rectangle area

Constraints: $1 \le n, m \le 1000$.

1162 — Sorting Methods

Given a permutation, independently find the minimum operations needed to sort it using: (1) adjacent swaps; (2) arbitrary swaps; (3) remove one element and insert it anywhere; (4) remove one element and insert it at the front. **In:** `n`; permutation p

Out: four minima in the order above

Constraints: $n \le 2 \cdot 10^5$. First answer needs 64-bit.

1188 — Bit Inversions

Maintain a bit string under m point flips. After each flip, report the maximum length of a contiguous run consisting entirely of equal bits. **In:** bit string s; `m`; flip positions $x_1 \ldots x_m$

Out: m run lengths, after each flip

Constraints: $|s|, m \le 2 \cdot 10^5$.

1191 — Cyclic Array

Partition a positive cyclic array into the minimum number of nonempty contiguous cyclic segments such that every segment sum is at most k. A feasible partition is guaranteed. **In:** `n k`; $x_1 \ldots x_n$

Out: minimum number of segments

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$, $k \le 10^{18}$; $x_i \le k$.

1670 — Swap Game

A 3×3 grid contains $1, \ldots, 9$ once each. Swap horizontally/vertically adjacent cells; find the minimum swaps to reach row-major order $1, 2, \ldots, 9$. **In:** three rows of three integers

Out: minimum number of adjacent swaps

1747 — Pyramid Array

Given n distinct values, use adjacent swaps to obtain an array that is first strictly increasing and then strictly decreasing; either part may be empty, so a fully monotone array is allowed. Minimize swaps. **In:** `n`; $x_1 \ldots x_n$

Out: minimum adjacent swaps

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$. Use 64-bit.

2186 — Special Substrings

Let C be the set of characters occurring anywhere in the input string. Count substrings in which every character in C occurs the same number of times. **In:** lowercase string s

Out: number of special substrings

Constraints: $|s| \le 2 \cdot 10^5$. Use 64-bit.

2414 — List of Sums

An unknown list A has n positive integers. The input multiset B contains every unordered pair sum $a_i + a_j$ for $i < j$; reconstruct any valid A. Existence is guaranteed. **In:** `n`; then $n(n-1)/2$ pair sums

Out: any valid n values of A, in any order

Constraints: $3 \le n \le 100$; values of A are positive and at most 10^9.

2422 — Multiplication Table

For odd n, consider the multiset $\{ij : 1 \le i, j \le n\}$ of the $n \times n$ multiplication table. Find its middle element after sorting all n^2 entries. **In:** `n`

Out: median table value

Constraints: $1 \le n < 10^6$, n odd. Use 64-bit arithmetic.

2425 — Stack Weights

Coins satisfy unknown strictly increasing weights $w_1 < w_2 < \cdots < w_n$. Process placements into left/right stacks; after each, determine whether the left stack is certainly heavier, the right is certainly heavier, or neither conclusion follows from the ordering alone. **In:** `n`; then n moves `c s`, $s \in \{1, 2\}$

Out: after each move: `>` left, `<` right, or `?`

Constraints: $n \le 2 \cdot 10^5$.

2427 — Letter Pair Move Game

A length-$2n$ string has exactly $n-1$ `A`'s, $n-1$ `B`'s, and one adjacent pair of empty boxes. A move transfers any adjacent occupied pair into the current empty pair, preserving order. Reach a state with every `A` before every `B`. **In:** `n`; initial string over `A,B,.`

Out: any valid solution with $k \le 1000$ moves, or `-1` if impossible

Constraints: $1 \le n \le 100$; if solvable, a solution of at most $10n$ moves exists.

3150 — Distinct Values Sum

For every subarray $x_a, \ldots, x_b$, let $d(a, b)$ be its number of distinct values. Compute $\sum_{a \le b} d(a, b)$ over all subarrays. **In:** `n`; $x_1 \ldots x_n$

Out: the total distinct-value sum

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$. Use 64-bit.

CSES — Additional Problems I Reference

Continued. Modulo $10^9 + 7$ only where explicitly stated.

3151 — Bubble Sort Rounds I

Bubble sort scans left-to-right each round, swapping adjacent out-of-order pairs as encountered. Given an arbitrary integer array, determine how many rounds are needed until it is sorted. **In:** `n`; $x_1 \dots x_n$

Out: number of bubble-sort rounds

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$.

3152 — Bubble Sort Rounds II

Using the same left-to-right bubble-sort round, find the entire array after exactly k rounds; k may be much larger than n. **In:** `n k`; $x_1 \dots x_n$

Out: array after k rounds

Constraints: $n \le 2 \cdot 10^5$, $0 \le k \le 10^9$, $x_i \le 10^9$.

3169 — Counting LCM Arrays

For each test, count positive-integer arrays $a_1, \dots, a_n$ such that $\operatorname{lcm}(a_i, a_{i+1}) = k$ for every adjacent pair. **In:** `t`; then tests `n k`

Out: answer for each test mod $10^9 + 7$

Constraints: $t \le 1000$, $2 \le n \le 10^9$, $1 \le k \le 10^9$.

3175 — Beautiful Permutation II

Construct the lexicographically smallest permutation of $1, \dots, n$ such that no adjacent values differ by 1. If none exists, report failure. **In:** `n`

Out: lexicographically minimum beautiful permutation, or `NO SOLUTION`

Constraints: $1 \le n \le 10^6$.

3190 — Distinct Values Splits

Split the whole array into one or more contiguous segments so that every segment contains pairwise distinct values. Count all possible choices of cut positions. **In:** `n`; $x_1 \dots x_n$

Out: number of valid splits mod $10^9 + 7$

Constraints: $n \le 2 \cdot 10^5$, $x_i \le 10^9$.

3193 — Square Subsets

Count subsets of the given multiset whose element product is a perfect square. The empty subset is included and has product 1. **In:** `n`; $x_1 \dots x_n$

Out: number of square-product subsets mod $10^9 + 7$

Constraints: $n \le 5000$, $1 \le x_i \le 5000$.

3213 — Water Containers Moves

Containers have capacities a, b and start empty. A move fills one container, empties one, or pours between them until the source empties or destination fills. End with exactly x units in A, minimizing the *total amount of water moved* across all moves. **In:** `a b x`

Out: minimum-water valid move sequence and its total, or `-1`

Constraints: $1 \le a, b, x \le 1000$; every printed move must transfer at least one unit.

3214 — Water Containers Queries

For many tests with capacities a, b, starting from both empty and using the same fill/empty/full-pour operations, decide whether it is possible to finish with exactly x units in container A. **In:** `t`; then tests `a b x`

Out: `YES` or `NO` for each test

Constraints: $t \le 1000$, $a, b, x \le 10^9$.

3294 — Subarray Sum Constraints

Construct integers $x_1, \dots, x_n$ satisfying every constraint (l, r, s): $x_l + \dots + x_r = s$. Output any solution with $|x_i| \le 10^{15}$, or detect inconsistency. **In:** `n m`; then m constraints `l r s`

Out: `YES` and any valid array, or `NO`

Constraints: $n \le 5000$, $m \le 2 \cdot 10^5$, $|s| \le 10^9$. Use 64-bit.

3301 — Maximum Average Subarrays

For every endpoint i, among all nonempty subarrays ending at i, choose one with maximum average; if several tie, choose the longest. Report its length. **In:** `n`; $x_1 \dots x_n$

Out: n optimal lengths, one for each endpoint

Constraints: $n \le 2 \cdot 10^5$, $1 \le x_i \le 10^6$.

3302 — Subsets with Fixed Average

Count nonempty subsets whose arithmetic mean is exactly the given integer a. **In:** `n a`; $x_1 \dots x_n$

Out: number of qualifying subsets mod $10^9 + 7$

Constraints: $n \le 500$, $a, x_i \le 500$.

3306 — Nearest Campsites I

Given n reserved and m free grid points, distance is Manhattan distance. For each free point consider its nearest reserved point; output the maximum of these nearest distances. **In:** `n m`; n reserved points; m free points

Out: $\max_{q \in F} \min_{p \in R} \|p - q\|_1$

Constraints: $n, m \le 10^5$, coordinates $\le 10^6$.

3307 — Nearest Campsites II

For every free campsite, compute its Manhattan distance to the nearest reserved campsite, preserving the input order of free points. **In:** `n m`; n reserved points; m free points

Out: m nearest-reserved distances

Constraints: $n, m \le 10^5$, coordinates $\le 10^6$.

3361 — Two Array Average

Choose a nonempty prefix of each of two length-n arrays. Maximize the average of all selected elements combined, and output any pair of prefix lengths attaining the optimum. **In:** `n`; array a; array b

Out: optimal prefix sizes i, j

Constraints: $n \le 10^5$, $a_i, b_i \le 10^9$; checker tolerance 10^{-6} on achieved optimum.

3404 — Permutation Subsequence

Given two arrays that are permutations, find a longest common subsequence: values must appear in the same left-to-right order in both arrays, with gaps allowed. **In:** `n m`; permutation a; permutation b

Out: LCS length, then any corresponding sequence

Constraints: $n, m \le 2 \cdot 10^5$; $1 \le a_i \le n$, $1 \le b_i \le m$.